

HOMEWORK #5

Artificial Intelligence

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Problem 1. Consider N data points (x_j, y_j) , where the y_j s are generated from the x_j s according to the linear Gaussian model in Equation (20.5). Find the values of θ_1 , θ_2 , and σ that maximize the conditional log likelihood of the data.

Answer: according to the linear Gaussian model

$$P(y | x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y - (\theta_1 x + \theta_2))^2}{2\sigma^2}}$$

We have

$$L = \sum_{j=1}^N \log \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y_j - (\theta_1 x_j + \theta_2))^2}{2\sigma^2}} = -N(\log \sigma + \log \sqrt{2\pi}) - \sum_j \frac{(y_j - (\theta_1 x_j + \theta_2))^2}{2\sigma^2}$$

Taking the derivative with respect to θ_1 , θ_2 , σ and setting them to zero, we get

$$\begin{aligned} \frac{\partial L}{\partial \theta_1} &= -\sum_j \frac{x_j(y_j - (\theta_1 x_j + \theta_2))}{\sigma^2} = 0 \\ \frac{\partial L}{\partial \theta_2} &= -\sum_j \frac{(y_j - (\theta_1 x_j + \theta_2))}{\sigma^2} = 0 \\ \frac{\partial L}{\partial \sigma} &= -\frac{m}{\sigma} + \sum_j \frac{(y_j - (\theta_1 x_j + \theta_2))^2}{\sigma^3} = 0 \end{aligned}$$

and the solutions can be computed as

$$\begin{aligned} \theta_1 &= \frac{m(\sum_j x_j y_j) - (\sum_j y_j)(\sum_j x_j)}{m(\sum_j x_j^2) - (\sum_j x_j)^2} \\ \theta_2 &= \frac{1}{m} \sum_j (y_j - \theta_1 x_j) \\ \sigma^2 &= \frac{1}{m} \sum_j (y_j - (\theta_1 x_j + \theta_2))^2 \end{aligned}$$

Problem 2. Consider a single Boolean random variable Y (the “classification”). Let the prior probability $P(Y = \text{true})$ be π . Let’s try to find π , given a training set $D = (y_1, \dots, y_N)$ with N independent samples of Y . Furthermore, suppose p of the N are positive and n of the N are negative.

1. Write down an expression for the likelihood of D (i.e., the probability of seeing this particular sequence of examples, given a fixed value of π) in terms of π , p , and n .
2. By differentiating the log likelihood L , find the value of π that maximizes the likelihood.
3. Now suppose we add in k Boolean random variables $X_1, X_2, \dots, X_k$ (the “attributes”) that describe each sample, and suppose we assume that the attributes are conditionally independent of each other given the goal Y . Draw the Bayes net corresponding to this assumption.
4. Write down the likelihood for the data including the attributes, using the following additional notation:
 - α_i is $P(X_i = \text{true} | Y = \text{true})$.
 - β_i is $P(X_i = \text{true} | Y = \text{false})$.
 - p_i^+ is the count of samples for which $X_i = \text{true}$ and $Y = \text{true}$.
 - n_i^+ is the count of samples for which $X_i = \text{false}$ and $Y = \text{true}$.
 - p_i^- is the count of samples for which $X_i = \text{true}$ and $Y = \text{false}$.
 - n_i^- is the count of samples for which $X_i = \text{false}$ and $Y = \text{false}$.

[Hint: consider first the probability of seeing a single example with specified values for $X_1, X_2, \dots, X_k$ and Y .]

5. By differentiating the log likelihood L , find the values of α_i and β_i (in terms of the various counts) that maximize the likelihood and say in words what these values represent.

Answer :

(a) The likelihood of D is $\pi^p(1 - \pi)^n$.

(b) Take the derivative of the log likelihood $L = \log(\pi^p(1 - \pi)^n)$ with respect to π and set it to zero:

$$\begin{aligned} \frac{\partial L}{\partial \pi} &= \frac{p}{\pi} - \frac{n}{1 - \pi} = 0 \\ \Rightarrow \pi &= \frac{p}{p + n} \end{aligned}$$

(c) The Bayes net corresponding to the assumption is as follows:

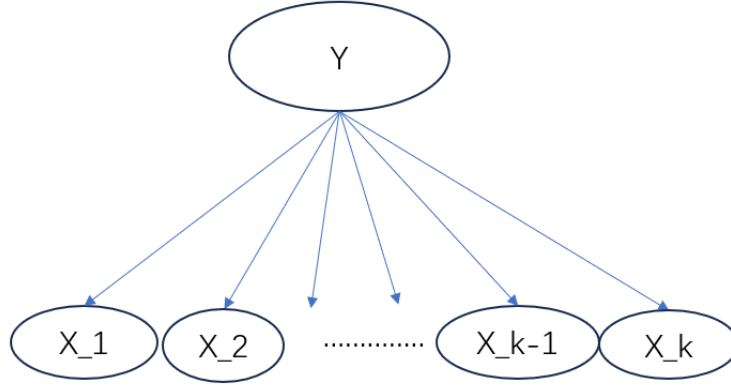


图 1 Bayes net for the attributes

(d) The likelihood for the data including the attributes is:

$$\begin{aligned}
 L &= \prod_i \pi^{p_i^+} \alpha_i^{p_i^+} \pi^{n_i^+} (1 - \alpha_i)^{n_i^+} (1 - \pi)^{p_i^-} \beta_i^{p_i^-} (1 - \pi)^{n_i^-} (1 - \beta_i)^{n_i^-} \\
 &= \pi^p (1 - \pi)^n \prod_i \alpha_i^{p_i^+} (1 - \alpha_i)^{n_i^+} \beta_i^{p_i^-} (1 - \beta_i)^{n_i^-}
 \end{aligned}$$

(e)

$$L = p \log \pi + n \log(1 - \pi) + \sum_i p_i^+ \log \alpha_i + n_i^+ \log(1 - \alpha_i) + p_i^- \log \beta_i + n_i^- \log(1 - \beta_i).$$

Take the derivative of the log likelihood L with respect to α_i and β_i and set it to zero:

$$\frac{\partial L}{\partial \alpha_i} = \frac{p_i^+}{\alpha_i} - \frac{n_i^+}{1 - \alpha_i} = 0 \quad \text{and} \quad \frac{\partial L}{\partial \beta_i} = \frac{p_i^-}{\beta_i} - \frac{n_i^-}{1 - \beta_i} = 0$$

giving $\alpha_i = p_i^+ / (p_i^+ + n_i^+)$ and $\beta_i = p_i^- / (p_i^- + n_i^-)$.

The values of α_i represent the fraction of cases where X_i is true given Y is true, and the values of β_i represent the fraction of cases where X_i is true given Y is false.

Problem 3.

Consider the following data set comprised of three binary input attributes (A_1 , A_2 , and A_3) and one binary output: Use the algorithm in Figure 18.5 (page 702) to learn a decision tree for these data. Show the computations made to determine the attribute to split at each node.

Example	A_1	A_2	A_3	Output y
x_1	1	0	0	0
x_2	1	0	1	0
x_3	0	1	0	0
x_4	1	1	1	1
x_5	1	1	0	1

Answer :

first we calculate the information gain of each attribute

$$G(A_1) = B\left(\frac{2}{5}\right) - \left(\frac{1}{5}B(0) + \frac{4}{5}B\left(\frac{2}{4}\right)\right) \simeq 0.971 - 0.800$$

$$G(A_2) = B\left(\frac{2}{5}\right) - \left(\frac{3}{5}B(0) + \frac{2}{5}B\left(\frac{2}{3}\right)\right) \simeq 0.971 - 0.551$$

$$G(A_3) = B\left(\frac{2}{5}\right) - \left(\frac{2}{5}B\left(\frac{1}{2}\right) + \frac{3}{5}B\left(\frac{1}{3}\right)\right) \simeq 0.971 - 0.951$$

Thus, we first split on A_2 . For $A_2 = 0$, it classifies $Y=0$. For $A_2 = 1$, we have remain (x_3, x_4, x_5) to classify. We calculate the information gain of A_1 and A_3 .

$$G(A_1) = B\left(\frac{2}{3}\right) - \left(\frac{1}{3}B(0) + \frac{2}{3}B\left(\frac{2}{2}\right)\right) \simeq 0.918 - 0.000$$

$$G(A_3) = B\left(\frac{2}{3}\right) - \left(\frac{1}{3}B(1) + \frac{2}{3}B\left(\frac{1}{2}\right)\right) \simeq 0.918 - 0.667$$

Thus, we split on A_1 for $A_1 = 0$, it classifies $Y=0$. For $A_1 = 1$, it classifies $Y=1$.

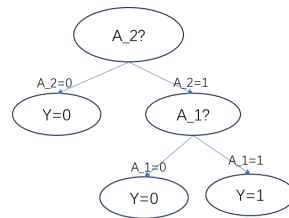


图 2 the decision tree

Problem 4. Suppose you had a neural network with linear activation functions. That is, for each unit the output is some constant c times the weighted sum of the inputs.

- Assume that the network has one hidden layer. For a given assignment to the weights $\mathbf{w}$, write down equations for the value of the units in the output layer as a function of $\mathbf{w}$ and the input layer $\mathbf{x}$, without any explicit mention of the output of the hidden layer. Show that there is a network with no hidden units that computes the same function.
- Repeat the calculation in part (a), but this time do it for a network with any number of hidden layers.
- Suppose a network with one hidden layer and linear activation functions has n input and output nodes and h hidden nodes. What effect does the transformation in part (a) to a network with no hidden layers have on the total number of weights? Discuss in particular the case $h \ll n$.

Answer :

(a) Assume the Input is I_k , the output of the hidden layer is H_j , and the output of the output layer is O_i . the weights are $w_{k,j}$ and $w_{j,i}$, respectively.

Assume the activation function for the hidden layer is $f_j(x) = a_jx + b_j$ and the activation function for the output layer is $g_i(x) = c_ix + d_i$.

we can calculate the output of the hidden layer as:

$$H_j = f_j(\sum_k w_{k,j} I_k) = a_j(\sum_k w_{k,j} I_k) + b_j$$

and the output of the output layer as:

$$\begin{aligned} O_i &= g_i(\sum_j w_{j,i} H_j) = c_i(\sum_j w_{j,i} H_j) + d_i \\ &= c_i(\sum_j w_{j,i} (a_j(\sum_k w_{k,j} I_k) + b_j)) + d_i \\ &= c_i(\sum_j w_{j,i} a_j(\sum_k w_{k,j} I_k)) + c_i(\sum_j w_{j,i} b_j) + d_i \\ &= c_i(\sum_k (\sum_j w_{j,i} a_j w_{k,j}) I_k) + c_i(\sum_j w_{j,i} b_j) + d_i \end{aligned}$$

we can see that the output of the output layer is a linear function of the input layer. Thus, we can compute the same function using a network with no hidden units with the following weights:

$$\begin{aligned} w_{k,i} &= \sum_j w_{j,i} a_j w_{k,j} \\ g_i(x) &= c_i x + c_i(\sum_j w_{j,i} b_j) + d_i \end{aligned}$$

(b) The calculation in part (a) can be extended to a network with any number of hidden layers.

For example, consider a network with n layers, we assume the output of $(n-2)$ th hidden layers is H_k^{n-2} , the output of $(n-1)$ th hidden layer is H_j^{n-1} , and the output of the output layer is O_i . Then, we can use the same method in (a) to calculate the output of the output layer. As (a) shows, we can join $(n-2)$ th hidden layer and $(n-1)$ th hidden layer into a single layer. Then the network can be simplified to a network with $(n-1)$ layers.

Finally, according to the induction principle, we can conclude that the network with any number of hidden layers can be simplified to a network with no hidden layers.

(c) The number of weights in the original network is $n \cdot h + h \cdot n = 2nh$.

The number of weights in the new network is $n \cdot n = n^2$.

Thus, if $h \ll n$, the number of weights in the new network is much larger than that in the original network. In this case, we can conclude that the transformation from a network with one hidden layer to a network with no hidden layers increases the number of weights significantly.